



Regenerative management and fertiliser regime impacts on phosphorus cycling in a temperate grazed pasture

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Abstract Phosphorus (P) cycling underpins productivity and fertiliser use efficiency in grazed pasture systems. However, the effects of regenerative practices such as enhancing pasture diversity and alternative grazing management remain poorly quantified. We investigated regenerative and conventional system effects on soil-plant-animal P dynamics in a three-year dryland grazed experiment. The regenerative system comprised a 12-species diverse pasture mix with short grazing periods and extended recovery, while the conventional system used best-practice grazing of *Medicago sativa* L. Both systems also included high and low P fertiliser treatments. Plant-available and biological soil P pools, acid phosphatase activity, herbage P concentrations, animal P intake and dung P returns were quantified. The P fertiliser treatments resulted in contrasting soil labile P pools, but management type had no effect. Soil microbial biomass P was greater under high compared to low fertiliser by the final sampling in Year 3 (24.4 vs. 15.8 mg P kg⁻¹, $P=0.001$). Averaged across P fertiliser treatments, soil acid phosphatase activity was 21% higher under regenerative management at the final soil sampling

($P=0.01$). Herbage P concentrations ranged from 1.4 to 4.4 g P kgDM⁻¹ and were strongly correlated with soil moisture ($R^2 = 0.50\text{--}0.65$). Animal P intake was 8.1 kg P ha⁻¹ higher in the conventional in Year 1 only. Dung P returns shifted from higher in the conventional system in Year 1 (+7 kg P ha⁻¹; $P<0.001$) to higher returns in the regenerative system in Year 3 (+3 kg P ha⁻¹; $P=0.03$). These results indicated that three years provided sufficient time for regenerative grazing management to begin measurably shifting P cycling processes, specifically in terms of P mineralisation by acid phosphatase activity and P returns via dung. However, bulk soil P fertility pools remained governed by fertiliser history rather than management type.

Keywords *Medicago sativa* L. · Multispecies pastures · Phosphorus fertiliser · Sheep grazing

Introduction

Phosphorus (P) is a major nutrient that is vital for plant growth and, therefore, productivity in pastoral systems (Gillingham et al. 1998). Because P is essential for high plant growth and development, it is typically applied as a fertiliser to maintain productivity (Haygarth et al. 2013). However, the amount of P required in pastoral systems remains debated, as it is influenced by plant species and root characteristics (Caradus 1980; Gillingham et al. 1998; Richardson

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et al. 2009), alongside differences in yield targets, soil properties, sampling depth and climatic conditions (Jalali et al. 2025). This uncertainty around optimal P management, combined with growing pressure to reduce input costs and environmental losses has driven interest in alternative pastoral management strategies, including regenerative agriculture.

Regenerative agriculture is a farmer-led, global movement that aims to improve the resilience and sustainability of food production systems in the face of climate change and environmental degradation (Giller et al. 2021; Schreefel et al. 2020). Regenerative practices can include: reduced or no tillage, multispecies cover crops, rotational grazing, reduced synthetic fertiliser and pesticide inputs (Khangura et al. 2023). Collectively, these practices aim to enhance soil function, with a particular emphasis on improving nutrient cycling, thereby reducing nutrient losses and external inputs while maintaining or enhancing productivity. Evidence suggests regenerative agriculture practices may enhance soil biological activity and herbage nutrient concentrations (Montgomery et al. 2022; Singh et al. 2023). However, outcomes for plant productivity and nutrient cycling remain variable and context dependent (Jordon et al. 2022a; Liebig and Doran 1999; Nguyen et al. 1995).

One of the main differences in regenerative agriculture practices compared to conventional is the use of fertilisers permitted in organic farming, such as reactive phosphate rock (RPR) (Condrón et al. 2000; Rowarth et al. 2020), which are typically slow-release and can temporarily reduce productivity relative to water soluble fertilisers before matching yields over time as the fertiliser dissolves (Bolan et al. 1990; Bolan & Hedley 1998; Obasoro et al. 2025). Regenerative systems also rely on diverse pasture species mixtures as a strategy for improving nutrient efficiency, since plant species vary in their ability to acclimate to low-P environments through differences in P utilisation efficiency and root traits such as architecture, root hair length and P “mining” characteristics (Haling et al. 2016; Hill et al. 2006; Simpson et al. 2014). At the same time, grazing management influences P cycling directly where grazing has been reported to increase root nutrient uptake and herbage P concentration relative to ungrazed systems (Chaneton et al. 1996), and dung can return up to 95% of ingested P to pasture (Safley et al. 1984). More broadly, P losses from grazed

systems are typically low. At Winchmore, for example, 94% of applied P (188 kg SSP ha⁻¹) was recovered in the soil-plant-animal system rather than lost (Nguyen and Goh 1992). The return of dung in these grazed systems can increase soil microbial biomass (Hatch et al. 2000; Rayne and Aula 2020), which strongly influences soil P cycling through nutrient transformations and storage (Rice et al. 1997), and mediates P availability to plants (Richardson and Simpson 2011; Schachtman et al. 1998), partly via extracellular enzymes such as acid phosphatase, which hydrolyses organic P into plant and microbial-available P (Dick 1994; Margalef et al. 2017; Piotrowka-Długosz 2020).

Despite these proposed mechanisms, the claim that diverse pasture mixtures enhance nutrient availability and uptake (Jordon et al. 2022b; Liebig and Doran 1999; Nguyen et al. 1995) has rarely been tested in grazed pasture systems. Instead, most uptake and nutrient availability studies have been completed in simple pastures or single species glasshouse experiments (Moir et al. 2016; Nguyen and Goh 1992; Rowarth et al. 1988). In regenerative pasture systems with the combination of diverse pasture species and alternative management, the effect on P cycling remains unclear (Rowarth et al. 2020), and larger scale experiments involving livestock capable of testing it have, until recently, not existed. Therefore, a pastoral grazed experiment was established in Canterbury, New Zealand. To date, in this experiment, pasture yield and botanical composition were quantified for the first 3.5 years. Watson et al. (2026b) reported that pasture yield was similar between regenerative and conventional pastures, with limited P fertiliser yield response in both systems. The lack of soil P fertiliser effect on yield was also reported by Jones et al. (2025) in the conventional treatment only. This result was unusual, as the low fertiliser treatment resulted in an Olsen P of 10 mg P kg⁻¹ which is reported to limit pasture yield in sedimentary soils (Edmeades et al. 2006). Understanding the key pools and movements of P in each system may provide valuable insights into how these systems compensate for P fertiliser inputs, or whether we are likely to observe yield differences in the future due to P depletion or accumulation. By quantifying and measuring changes in the most labile pools of soil P (microbial biomass P and plant-available P), as well as the intake of P by grazing animals and its return in dung, we may determine

key changes in P pools and whether regenerative management impacts P cycling.

To address this, we quantified P cycling across soil, plant and animal components within the same regenerative and conventional grazing systems, under high and low soil P fertiliser inputs, over a three year period. The objective of this study was to quantify and compare key components of P cycling in both regenerative and conventional temperate pasture systems under contrasting P fertiliser inputs. We hypothesised that soil P concentrations and herbage uptake would increase while acid phosphatase activity would decrease with increased P fertilisation. Also, in the regenerative system, the amount of labile P in the system may decrease due to the use of poorly soluble fertiliser (RPR), but acid phosphatase activity and microbial biomass would increase.

Materials and methods

Experimental site

To assess the potential benefits or drawbacks of regenerative compared to conventional agriculture in New Zealand, “Whenua Haumanu, nurturing the land through exploring pastoral farming” was established in 2022. The Whenua Haumanu project includes the Regenerative Agriculture Dryland Experiment (RADE) located at the Field Research Centre at Lincoln University, New Zealand (43°38'54.20"S, 172°27'34.30"E, 9 m above sea level). The soil was classified as Templeton silt loam (Typic Immature Pallic New Zealand classification; Udic Ustochrept, USDA classification). The experiment included 20 replicates of four treatments: high fertiliser conventional (HC), high fertiliser regenerative (HR), low fertiliser conventional (LC) and low fertiliser regenerative (LR). A total of 80 grazed plots were arranged across a 7.7 ha area in a five-block Latin square design to control for spatial variation, with each of the four treatments represented once in every row and column within each square. A map was included in Watson et al. (2024).

This manuscript focuses on the first two squares of the experiment sown in December 2021: plots 1–16 (Square 1) and plots 17–32 (Square 2). The 32 plots were 29 × 30 m each and included eight replicates of the four treatments (four in each square). The

regenerative pasture was a 12-species pasture mixture (as decided by the technical advisory group of farmers outside of the experiment), including prairie grass (*Bromus willdenowii* L.), chicory (*Cichorium intybus* L.), cocksfoot (*Dactylis glomerata* L.), tall fescue (*Lolium arundinaceum* (Schreb.) Darbysh.), meadow fescue (*Lolium pratense* (Huds.) Darbysh.), lucerne, phalaris (*Phalaris aquatica* L.), timothy (*Phleum pratense* L.), plantain (*Plantago lanceolata* L.), red clover (*Trifolium pratense* L.), white clover (*Trifolium repens* L.) and subterranean clover (*Trifolium subterraneum* L.). In contrast, the conventional treatments were a monoculture of lucerne. Lucerne was selected for superior efficiency in dry matter (DM) yield and quality over grass/clover mixes, particularly under hot and dry summer conditions (Mills et al. 2015, 2008; Moot et al. 2008). These lucerne-based pastures are vital for productivity in dry east coast regions in New Zealand have largely contributed to improved ewe performance in sheep and beef systems in the past 30 years (Moot 2022). Therefore, comparisons between lucerne and multispecies pastures may be more meaningful than those involving drought-sensitive grass and clover systems. The main differences in treatment between regenerative and conventional systems were pasture species diversity and grazing management. However, other differences include; no herbicide or pesticide application in the regenerative and the use of organic P fertiliser after the initial soil P development phase. Further details on the entire RADE were summarised by Robb et al. (2025).

For this site, weather data were collected from Broadfields weather station, approximately 2.5 kms north of the experimental site. Total rainfall was 811 mm, 544 mm and 857 mm for Year 1 (July 2022–June 2023), Year 2 and Year 3, respectively (Watson et al. 2026b). The mean rainfall for the weather station was 627 mm between 1972 and 2023. Throughout the experimental period, the average maximum air temperature was 17.7 °C (range = 6.8–33.6 °C), and the average minimum air temperature was 7.7 °C (range = −4–21.2 °C).

Site management

Grazing management

The regenerative grazing management aims to replicate natural grazing systems, where dense livestock

herds are rapidly rotated through multiple paddocks with brief grazing periods and extended recovery times (Morris 2021). For conventional treatments, the sheep were divided into two groups of 10 (two for each high fertiliser conventional and low fertiliser conventional) and grazed each plot for 4–6 days. The exact number of grazing days per plot in both systems depended on pasture availability, but conventional plots always had twice the grazing duration of regenerative plots. Once assigned to one of the four treatments, the sheep remained within their treatment and grazed all plots within it across the five blocks. For further details, see Watson et al. (2026b). The treatments were also periodically mowed/topped post-grazing, with the clippings returned to the pasture. For the conventional, the post-grazing mowing occurred in late summer each year to maintain pasture quality. For the regenerative, the post-grazing mowing was only implemented in Year 3 onwards, upon consultation with the technical advisory group of farmers to maintain pasture quality through late summer.

Fertiliser treatment

The P fertiliser treatments in this experiment were either high or low soil P status. The low fertiliser (LF) plots had a target Olsen P of 10 mg P kg⁻¹ while the high fertiliser (HF) plots had a target of 20 mg P kg⁻¹ as recommended by Fertiliser Association of New Zealand (2018). Before sowing, single superphosphate (SSP; 9% P, 11% S) was applied to the surface of the HF plots at a rate of 36 kg P ha⁻¹ (Watson et al. 2026b). In spring 2023 (mean date, 11 October), SSP was surface applied (HF: 54 kg P ha⁻¹ and LF: 5 kg P ha⁻¹). Some exceptions were made for plots that required higher fertiliser inputs to reach the Olsen P target of 20 mg P kg⁻¹. Post fertiliser application in spring 2023, Olsen P has remained at an average of 22 mg P kg⁻¹ for HF and 10 mg P kg⁻¹ for LF. Elemental sulphur (S90, 90% S) was applied to maintain adequate concentrations for plant production in all treatments (105 kg S ha⁻¹). In spring 2024, reactive phosphate rock (RPR; 11% P, 28% calcium) was surface applied to the HF regenerative treatment (14 kg P ha⁻¹) to replace SSP, aligning with regenerative farming practices. The HF conventional treatment received a surface application of SSP at a rate of 14 kg P ha⁻¹ at the same time. Elemental S was

again applied (18 kg S ha⁻¹) in spring 2024 to all treatments.

Soil pH between spring 2022 and autumn 2025 was a mean of 6.5 (range = 7.2–5.6, across all treatments and seasons). Total soil C and N averaged 2.06 ± 0.022% and 0.14 ± 0.0017% across all treatments between spring 2022 and spring 2024. This resulted in a mean soil C:N ratio of 14.5. Anion storage capacity was reported by DeLucca-Agrelo et al. (2026) at a site ~0.5 km east of the experiment at 25%. This anion storage capacity is considered low and refers to the soils ability to retain anions such as P (Reid et al. 2020).

Soil sampling and analysis

Soil samples were collected in spring (October 2023, October 2024) and autumn (May 2024, April 2025). From each plot, 20 cores (25 mm diameter, 75 mm depth) were taken along a diagonal transect from the north-east to the south-west corner and bulked. Samples were stored at 4 °C during processing, homogenised by hand, and sieved to < 4 mm. A sub-sample of fresh soil was separated for microbial biomass P measurement. For enzyme analysis, another sub-sample was further sieved moist to 2 mm and frozen at -20°C (< 6 months) before analysis. The remaining soil was air-dried at 30°C and sieved to 2 mm for pH, Olsen P, resin P, and Mehlich P analysis. Soil pH was measured in a 1:2.5 soil-to-water suspension. Air-dried soil (10 ± 0.05 g) was mixed with 25 mL deionised water, shaken for 30 min, and left to stand for 4–24 h (Blakemore et al. 1987).

Three plant-available soil P soil extractions were used to measure differences in the most labile soil P pool. These three tests were examples of acid, neutral, and alkaline extractions. The Mehlich test is the most effective at extracting P as the extractant has a low pH (2.5), which dissolves less available forms of P (Mehlich 1984). In contrast, resin P is a neutral extraction with a strip of anion exchange membrane (Saggar et al. 1990a). Resin P was reported to correlate well with pasture yield and accounted for 76% of the variation in pasture yield compared with only 42% for Olsen P (Saggar et al. 1999). The Olsen P test remains the standard labile P test in New Zealand as it was developed earlier than resin P as a basic extraction using sodium bicarbonate at pH 8.5 (Olsen et al. 1954). Olsen P was developed

for alkaline soils, though many New Zealand soils are acidic, and some are formed in recent sedimentary material that contains apatite minerals that dissolve in acid tests such as Truog or Mehlich (Grigg 1977). Therefore, acid tests may overestimate plant-available P, and Olsen P became the adopted test.

Olsen P was determined using a modified method (Blakemore et al. 1987; Murphy & Riley 1962; Olsen et al. 1954). Briefly, 1 g of air-dried, 2 mm-sieved soil was extracted with 20 mL of 0.5 M NaHCO_3 (pH 8.5) by end-over-end shaking (30 min), followed by centrifugation (2000 rpm, 10 min) and filtration (Whatman 42). Extracts were stored frozen prior to analysis using a Smartchem 200 discrete analyser (AMS, Frepillon, France). Mehlich-3 P was determined using a modified method of Mehlich (1984); 2.5 g of air-dried, 2 mm-sieved soil was extracted with 25 mL of Mehlich-3 extracting reagent (pH 2.5) by end-over-end shaking (5 min), followed by filtration (Whatman 42). Phosphorus concentrations in the extracts were measured colourimetrically following Murphy and Riley (1962). The colourimetric measurements were determined using a UV/Vis spectrophotometer (Varian Cary 50, Varian, Victoria, Australia). Resin-extractable P was determined using a modified method of Olsen et al. (1954) and Saggar et al. (1990a), using VWR Chemicals BDH® resin strips pre-soaked in 0.5 M sodium bicarbonate. 1 g of air-dried, 2 mm-sieved soil was shaken with 30 mL of deionised water and one resin strip for 16 h. The resin was then eluted in 20 mL of 0.5 M NaCl for 30 min. Phosphorus concentrations in the extracts were measured colourimetrically using the same protocol as Mehlich P.

Microbial biomass P was measured using methods adapted from Grace et al. (2006) and Brookes et al. (1982). Field-moist soil (0.5 g air-dry equivalent, 4 mm sieved) was prepared in triplicate for non-fumigated, fumigated, and P-spiked sets. Fumigated samples were treated with 0.5 mL of chloroform, incubated for 24 h, and then evacuated under vacuum to remove residual chloroform. All samples were extracted with 7.5 mL of 0.5 M NaHCO_3 (pH 8.5); for spiked samples, the extract contained 25 $\mu\text{g P mL}^{-1}$. Extracts were shaken for 30 min, centrifuged (3500 rpm, 10 min), filtered (Whatman 42), and analysed for P using a Smartchem 200 discrete analyser (AMS, Frepillon, France).

Acid phosphatase enzyme activity was determined using methods adapted from Bell et al. (2013). Soil slurries were prepared for each soil sample using 2.75 g of moist soil (2 mm sieved). The soil was homogenised using a high-powered blender with 165 mL of 50 mM sodium acetate buffer adjusted to match the soil's average pH (6.59). The soil slurry was poured into glass bowls and stirred continuously at 200 rpm. 200 μL of each soil slurry and 50 μL of 4-Methylumbelliferone (MUB)-linked phosphatase substrate (0.2 mM) were added to Costar 96 Flat Bottom Black Polystyrene plates with 8 experimental replicates were soil sample. Concurrently, a standard plate with a range of MUB concentrations (0–0.2 mM) was prepared for each soil slurry in triplicate. The plates were then incubated for 2 h at 20°C. Following incubation, the plates were read on a fluorometer with a microplate reader (Infinite 200Pro, Tecan i-control) at an excitation of 365 nm and emission of 450 nm.

Dung phosphorus

Initial dung collection began in November 2023 (spring) and continued for ~5 weeks or until each plot had been grazed. The collection was repeated in January 2024 (summer), October 2024 (spring) and February 2025 (summer). Dung (~10 deposits) was collected from each plot immediately following grazing. The dung was then dried at 65°C for a minimum of 48 h. The dung was ground using a mechanical processor to create a homogenised sample. Dung samples (0.2 g) were then acid digested (using a modified version of EPA Method 200.2, Rev. 2.8) and analysed for total P content using an Inductively Coupled Plasma Optical Emission Spectrophotometer (ICP-OES: Varian 720-ES ICP-OES, Varian, Melbourne, Australia).

Herbage analysis

Herbage mass was measured for all plots before and after each sheep grazing (approximately monthly), for all grazing events on all plots using three evenly distributed 0.5 m² quadrats along a transect down the middle of the plot. Quadrats were cut to ~1 cm above ground level, and samples were weighed fresh. A representative sub-sample was dried at 65 °C for 48 h to determine DM content, and another subsample of ~200 plant pieces was separated by species (unsown

species grouped as weeds). For all pre-grazing pasture cuts, an additional subsample was retained for acid digestion (using a modified version of EPA Method 200.2, Rev. 2.8) and total P analysis, by Inductively Coupled Plasma Optical Emission Spectrophotometer (ICP-OES: Varian 720-ES ICP-OES, Varian, Melbourne, Australia). The pre-grazing herbage P concentration was measured across 680 grazing events across all plots ($n = 32$) and all years (July 2022–June 2025). The post-grazing pasture P concentration was measured on 132 samples (19% of grazings) across all treatments. The remaining post-grazing pasture samples were estimated for P concentration based on the relationship with pre-grazing P concentrations (Fig. S1).

Animal herbage P intake was calculated from the herbage mass measurements obtained by quadrat cuts. The intake was calculated based on pre-grazing pasture P minus post-grazing pasture P (~3–6 days following pre-grazing). Therefore, this measurement is essentially herbage P uptake accounting for utilisation as the quadrat cuts quantified the amount of herbage consumed by the animal. The refused pasture was assumed to be recycled in the system. Prior to 30 March 2022, post-grazing herbage mass was visually estimated. It was assumed that dead material was not consumed; therefore, dead material was removed from the herbage mass totals. When plots were grazed multiple times within a season, herbage intake was summed to obtain seasonal totals, which were then averaged across treatments. Plots 1–4 were cultivated in Year 3 and were excluded from analyses for the period 1 July 2023 to 30 June 2025 (Years 3 and 4).

Phosphorus cycle calculations

Dung phosphorus returns

Dung samples were collected during spring 2023, summer 2023, spring 2024, and summer 2024. For the remaining grazing events, dung P concentration was estimated from the relationship with pre-grazing pasture P concentration (Fig. S2). This was completed to determine yearly dung P returns. Urine P returns were not considered, as previous studies have determined that urine P concentrations are negligible

compared to dung P (Barrow and Lambourne 1962; Haynes and Williams 1993).

Dung P return per grazing event was calculated as:

$$\begin{aligned} \text{Dung P returns} & (\text{kg P ha}^{-1}) \\ &= \text{Adjusted sheep ha}^{-1} \\ &\quad \times (\text{Dung P concentration} (\text{g P kgDM}^{-1}) \times 0.1) \quad (1) \\ &\quad \times \text{Dung DM} (\text{kgDM sheep}^{-1} \text{d}^{-1}) \\ &\quad \times \text{Grazing duration} (\text{d}) \end{aligned}$$

Sheep stock units were adjusted by class (adjusted sheep ha^{-1}): ewe = 1.0, lamb = 0.3, hogget = 0.6, ram = 0.7. Typically, one 55 kg ewe feeding one lamb is classed as 1.0 stock unit, calculated on an energy requirement basis (Parker 1998).

Dung dry matter (DM) was calculated as:

$$\begin{aligned} \text{Dung DM} (\text{kgDM sheep}^{-1} \text{d}^{-1}) &= \text{DM intake} (\text{kgDM d}^{-1}) \\ &\quad \times (1 - \text{Digestibility}) \quad (2) \end{aligned}$$

Equation 2 was determined by rearranging the equation used by Bruckental et al. (1987), where DM intake = dung DM/(1 – digestibility). The results from this calculation were dung DM returns at 0.519 $\text{kgDM sheep}^{-1} \text{d}^{-1}$ for regenerative treatments and 0.457 $\text{kgDM sheep}^{-1} \text{d}^{-1}$ for conventional treatments using the Eqs. 3–5 below to determine DM intake and digestibility.

Daily DM intake was calculated using Eq. 3. Equation 4 was used to determine the ewe energy requirements:

$$\text{DM intake} (\text{kgDM d}^{-1}) = \frac{\text{Energy requirement} (\text{MJ ME d}^{-1})}{\text{ME of pasture} (\text{MJ ME kgDM}^{-1})} \quad (3)$$

Metabolizable energy (ME) values used were 9.5 MJ kg DM^{-1} (regenerative) and 10 MJ kg DM^{-1} (conventional) from Robb et al. (2025), resulting in DM intake of 1.27 $\text{kgDM ewe}^{-1} \text{d}^{-1}$ for regenerative and 1.21 $\text{kg DM ewe}^{-1} \text{d}^{-1}$ for conventional.

Energy requirement for each ewe was:

$$\begin{aligned} \text{Energy requirement} (\text{MJ ME d}^{-1}) &= 0.6 \times \text{LW}^{0.75} \\ &= 12.1 \text{ MJ d}^{-1} \quad (4) \end{aligned}$$

Ewe liveweight (LW) was between 50 and 60 kg (Robb et al. 2025). For this calculation, 55 kg was used.

Digestibility was calculated as:

$$\text{Digestibility} = \frac{\text{ME} - \text{losses}}{\text{Gross energy of pasture}} \quad (5)$$

Losses were assumed as 15% of ME, including urinary and methane (Morgavi et al. 2023). Gross energy of pasture was assumed to be 18.5 megajoules per kgDM⁻¹ (Yan et al. 2000). Resulting digestibility values were 59.1% (regenerative) and 62.2% (conventional).

Animal product removal

Animal losses via carcass were calculated using 4.5 g of P per kg of LW (Table S1), and losses via wool using 2 g P kg wool⁻¹ (Williams and Haynes 1992).

Statistical analysis

The data were analysed in R (R Core Team 2024) using RStudio (Posit Software PBC 2024) using linear mixed-effects models with an analysis of variance (ANOVA). The models included P fertiliser treatment, system and season/year or sampling date as fixed effects (along with their interactions). The season and/or date were treated as fixed rather than random effects because they represent specific, non-random conditions of interest in this experiment, and our objective was to estimate and compare treatment responses across the measured seasons/dates and their interactions with treatments. The random effects were the Latin square experiment design features, including square, column, row and plot, which account for

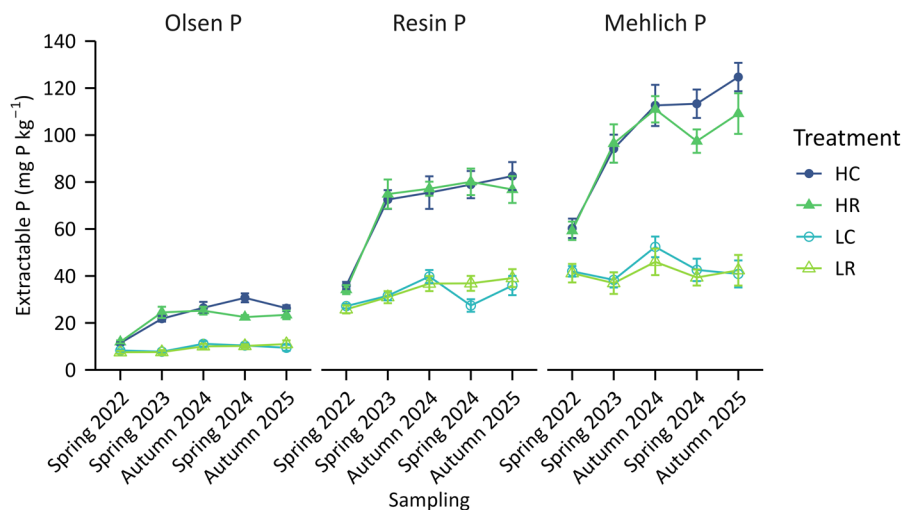
the hierarchical field structure and repeated observations within plots. Significant fixed effects and/or interactions from an ANOVA on the model were tested via a post hoc test with pairwise comparisons of estimated marginal means using Tukey-adjusted pairwise contrasts. Data were square root or log transformed where necessary to ensure equal variance and normality. Spearman's correlation analysis and principal component analysis (PCA) were completed to determine relationships between soil biological and biochemical properties. Spearman's rank correlation was performed to ensure the non-normal variables were accounted for. All the above analyses were conducted and the packages *lme4* (Bates et al. 2015) *lmerTest* (Kuznetsova et al. 2017), *emmeans* (Lenth 2024), *multcomp* (Hothorn et al. 2008) *car* (Fox & Weisberg 2019), *vegan* (Oksanen et al. 2020), *Hmisc* (Harrell Jr. and Dupont 2019) and *ggplot2* (Wickham 2016).

Results

Soil phosphorus

Fertiliser application in spring 2023 increased soil plant-available P concentrations in both high fertiliser regenerative (HR) and high fertiliser conventional (HC) treatments (Fig. 1). Since spring 2023, the low fertiliser (LF) treatments had mean Olsen P, resin P and Mehlich P of 9, 35 and 42 mg P kg⁻¹, respectively. Comparatively, in the high fertiliser

Fig. 1 Mean Soil Olsen P, Resin P and Mehlich P (mg P kg⁻¹) each season for high fertiliser conventional (HC; ●), high fertiliser regenerative (HR; ▲), low fertiliser conventional (LC; ○) and low fertiliser regenerative (LR; △) treatments. Error bars indicate ± 1 SEM (n = 7 or 8)



(HF) treatments, Olsen P, resin P and Mehlich P had means of 24, 77 and 107 mg P kg⁻¹, respectively. The difference in plant available soil P between regenerative and conventional management was observed only in the Olsen P measurement in spring 2024, where HC was higher than HR ($P=0.001$). For Mehlich and resin P, there were no differences in plant-available P between regenerative and conventional. Significant differences ($P<0.05$) between HF and LF treatments were observed in all seasons except spring 2022 across all plant-available P tests.

Seasonally, microbial biomass P fluctuated (Fig. 2a). In autumn 2025, microbial biomass P peaked at a mean of 20.1 mg P kg⁻¹ across all treatments, which was higher than in spring 2023 ($P<0.001$) and autumn 2024 ($P<0.001$). The increase in microbial biomass P between spring 2023 and autumn 2025 was 35, 91, 12 and 44% for HC, HR, LC and LR, respectively. The final autumn 2025 sampling was also the only sampling in which significant treatment differences were recorded. In that sampling, microbial biomass P was higher in HF than LF (24.4 vs. 15.8 mg P kg⁻¹, $P=0.001$).

Acid phosphatase activity was the same for all treatments across the first two sampling dates in spring 2023 and autumn 2024 (Fig. 2b). Between spring 2023 and autumn 2025, acid phosphatase activity increased by 43, 103, 58 and 90% for HC, HR, LC and LR, respectively. In autumn 2025, HR was higher than HC ($P=0.03$), and regenerative PHOS activity was, on average, 21% greater than conventional PHOS activity ($P=0.01$).

Soil moisture to 500 mm depth was not different among treatments within each season; however, strong seasonal variations were observed (Fig. 2c). In autumn 2024, soil moisture was lower than spring 2023 ($P<0.001$), spring 2024 ($P<0.001$) and autumn 2025 ($P=0.001$).

Herbage phosphorus

Pre-grazing herbage P concentrations varied substantially between seasons and treatments (Fig. S3). The herbage P concentration was lowest in autumn 2024 for LR (1.4 g P kgDM⁻¹) and highest in winter 2024 for HC (4.4 g P kgDM⁻¹). On average, the conventional treatment had higher herbage P

concentrations than the regenerative treatment (3.0 vs. 2.7 g P kgDM⁻¹; $P<0.001$). The HF treatment also had higher herbage P concentrations than LF (3.1 vs. 2.7 g P kgDM⁻¹; $P<0.001$). The herbage P concentration had a positive linear relationship with soil moisture content (Fig. S4). The highest correlation was observed for LR (slope = 0.08, $R^2 = 0.65$) and the lowest correlation was in HR (slope = 0.08, $R^2 = 0.50$).

Each spring, herbage P concentrations and pasture growth were higher, leading to higher animal P intake (Fig. 3). However, in spring, when intake was at its maximum, there was no difference in P intake between treatments ($P>0.05$). In summer 2023, autumn 2023, and 2025, the animal intake of P for the conventional treatments was higher than for the regenerative treatments by 6.1, 1.9, and 2.2 kg P ha⁻¹, respectively. The P fertiliser treatment resulted in higher animal intake in HF compared to LF in summer 2023, summer 2024, and spring 2024 by 0.9, 2.1 and 2.4 kg P ha⁻¹, respectively.

Yearly, calculated animal plant P intake was relatively consistent across treatments and years. Over the three-year period, conventional intake was on average 5.3 kg P ha⁻¹ yr⁻¹ higher than regenerative ($P<0.001$). Within years, the Year 1 conventional intake was 8.1 kg P ha⁻¹ greater than regenerative ($P<0.001$). In Years 2 and 3, there was no difference due to system or fertiliser ($P>0.05$).

Dung phosphorus returns

Dung P concentrations were higher in the conventional treatments than the regenerative treatments on average across all sampling periods (Fig. 4). Dung P concentrations were also higher in the HF than the LF treatments (9.0 vs. 7.8 g P kgDM⁻¹, $P<0.001$). In spring 2023, conventional treatments had higher dung P concentrations (10.3 vs. 6.6 g P kgDM⁻¹, $P<0.001$) than regenerative treatments. Dung P concentration was higher in the HF treatments compared to LF in spring 2023 ($P<0.003$), spring 2024 ($P=0.009$), and summer 2025 ($P=0.01$). In all treatments, dung P concentration was weakly, but positively, correlated with herbage P concentration (Fig. S2). The highest correlation was observed in the LR treatment (slope = 2.46, $R^2 = 0.31$), and the lowest in the LC treatment (slope = 1.3, $R^2 = 0.23$).

Fig. 2 Soil microbial biomass P (mg P kg^{-1}), phosphatase activity ($\text{nmol activity g dry soil}^{-1} \text{ h}^{-1}$) and soil moisture (%) to 500 mm. Treatments were high fertiliser conventional (HC), high fertiliser regenerative (HR), low fertiliser conventional (LC), and low fertiliser regenerative (LR) represented by blue for conventional and green for regenerative with stripes for high fertiliser and none for low fertiliser. Values are means ± 1 standard error ($n = 7$ or 8). Bars that share the same letters are statistically the same ($P > 0.05$) within each season

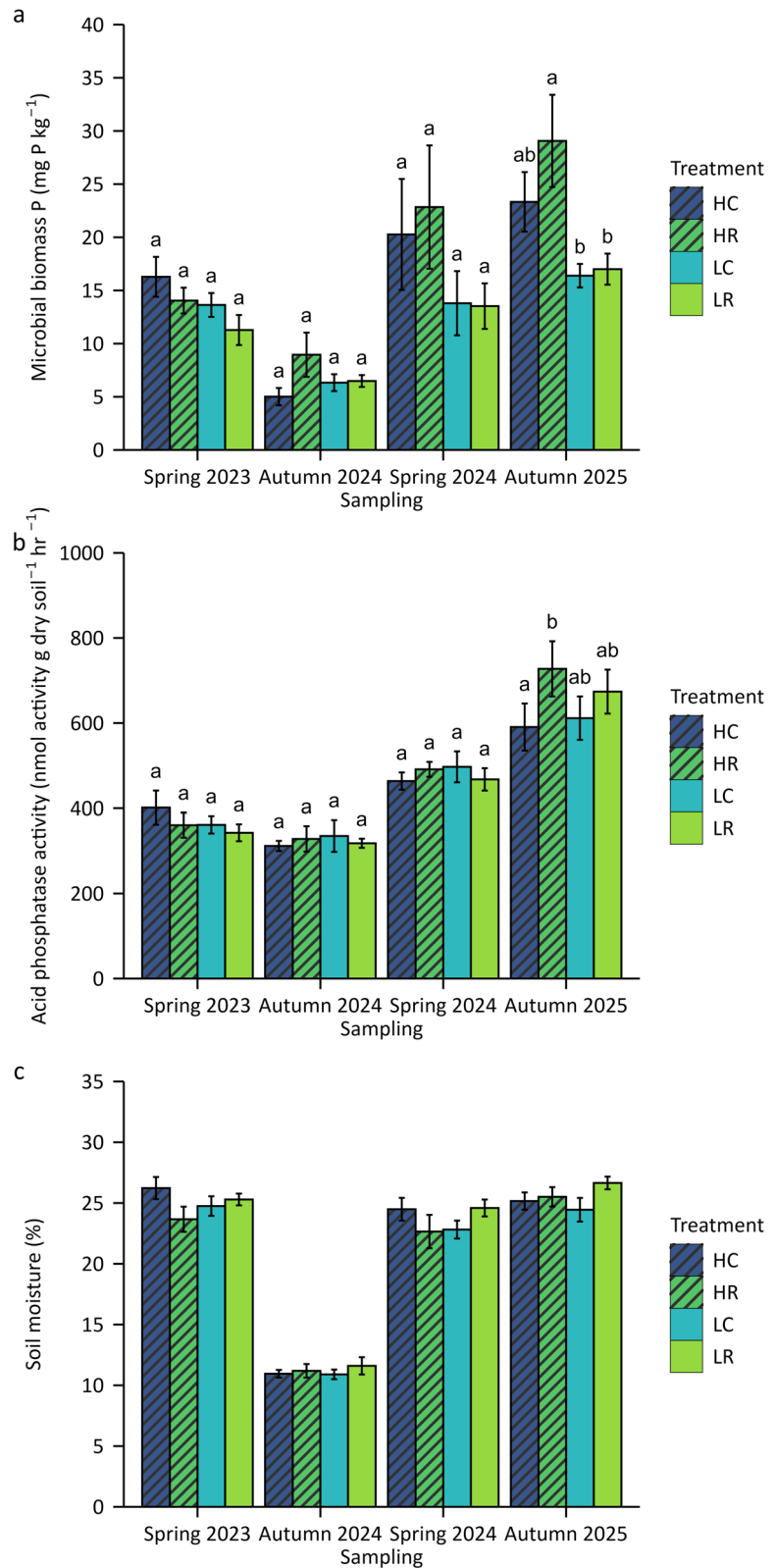


Fig. 3 Mean animal P intake (kg P ha^{-1}) per season for high fertiliser conventional (HC), high fertiliser regenerative (HR), low fertiliser conventional (LC), and low fertiliser regenerative (LR), represented by blue for conventional and green for regenerative and stripes for high fertiliser and none for low fertiliser. Values are means ± 1 standard error ($n = 7$ or 8). Bars sharing the same letter are not statistically different ($P > 0.05$) within each year

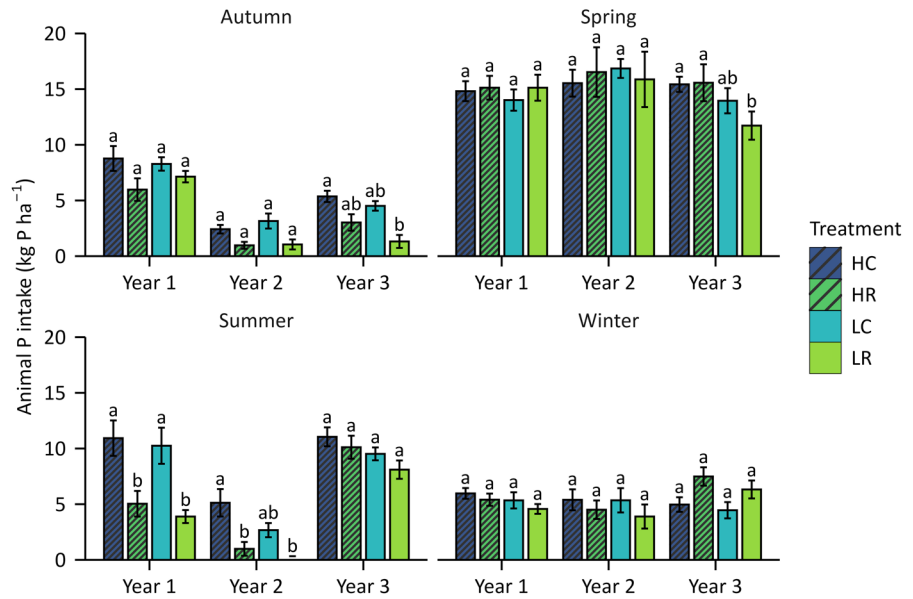
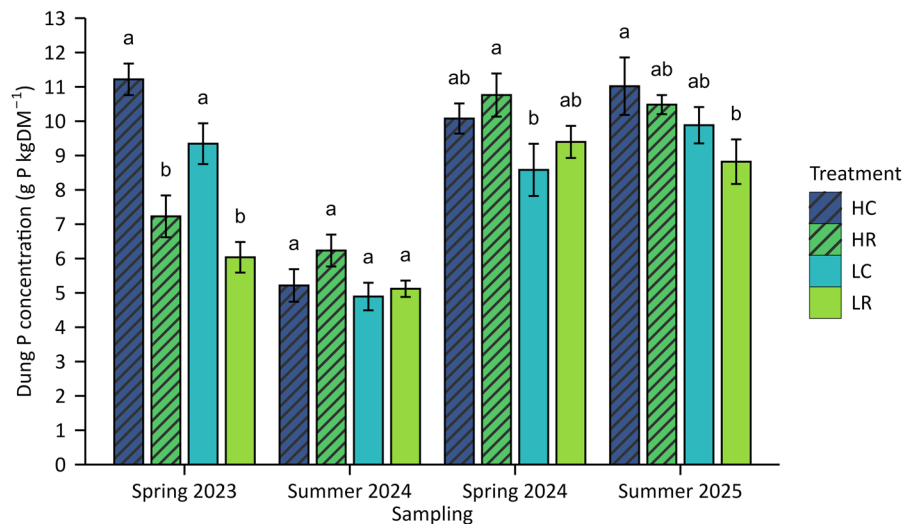


Fig. 4 Dung P concentrations (g P kgDM^{-1}) of samples collected in spring and summer over two years for high fertiliser conventional (HC), high fertiliser regenerative (HR), low fertiliser conventional (LC), and low fertiliser regenerative (LR) treatments, represented by blue for conventional and green for regenerative and stripes for high fertiliser and none for low fertiliser. Values are means ± 1 standard error ($n = 7$ or 8). Bars sharing the same letter are not statistically different ($P > 0.05$) within each year



The calculated dung P returns were higher in the conventional treatments than in the regenerative treatments in Year 1 by $7 \text{ kg P ha}^{-1} \text{ yr}^{-1}$ (Fig. 5; $P < 0.001$) and Year 2 by $4 \text{ kg P ha}^{-1} \text{ yr}^{-1}$ ($P = 0.002$). However, in Year 3, dung P returns were higher in the regenerative treatments than in the conventional treatment by 3 kg P ha^{-1} ($P = 0.03$). Only in Year 3 did P fertiliser influence dung P returns, with the HF treatments returning $6 \text{ kg P ha}^{-1} \text{ yr}^{-1}$ more than LF ($P < 0.001$).

Relationship between key phosphorus cycling components and soil conditions

On the basis that key P cycle components and soil conditions were correlated with each other (Table S1), a PCA was conducted to reveal that the data were divided into two principal components (Fig. 6). The first component explained 31% of the data variability and was mainly described by soil moisture (loading = 0.54), microbial biomass P (loading = 0.52) and acid phosphatase (loading = 0.45). The second

Fig. 5 Dung P returns (kg P ha⁻¹) each year for high fertiliser conventional (HC), high fertiliser regenerative (HR), low fertiliser conventional (LC), and low fertiliser regenerative (LR) treatments, represented by blue for conventional and green for regenerative and stripes for high fertiliser and none for low fertiliser. Values are means $\pm$ 1 standard error (n = 7 or 8). Bars sharing the same letter are not statistically different ($P > 0.05$) within each year

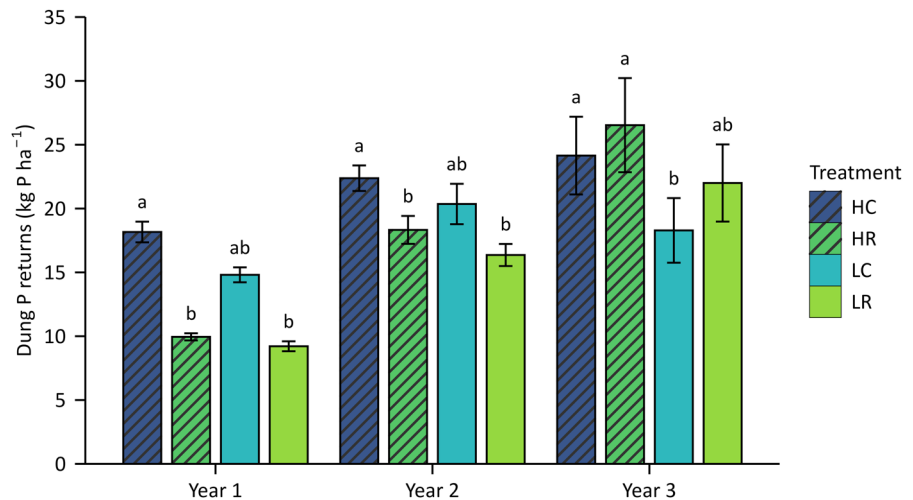
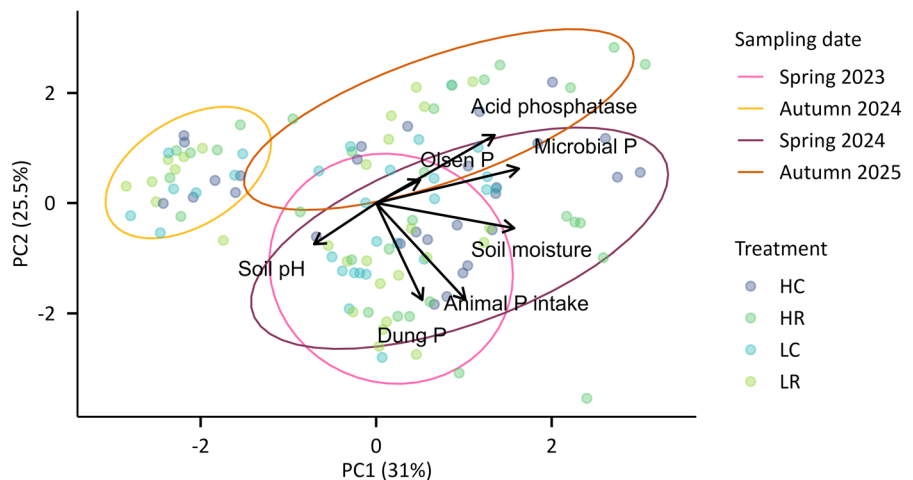


Fig. 6 Principal component analysis (PCA) of key phosphorus cycling components and soil conditions by treatment (points): high fertiliser conventional (HC, dark blue), high fertiliser regenerative (HR, dark green), low fertiliser conventional (LC, light blue), and low fertiliser regenerative (LR, light green), and sampling date (ellipses): spring 2023 (pink), autumn 2024 (yellow), spring 2024 (purple) and autumn 2025 (orange)



principal component explained 25.5% of the data variability and was mainly described by dung P (loading = -0.59), animal P intake (loading = -0.58) and acid phosphatase (loading = 0.41). The PCA revealed that the sample groups were largely driven by sampling date rather than treatment. While the points representing different treatments were relatively evenly distributed across the ordination space, the ellipses for different seasons were clearly separated, particularly for autumn samplings (Fig. 6).

Discussion

Phosphorus cycling in pastoral systems is vital for agricultural productivity and the efficiency of

fertiliser use (Haynes and Williams 1993). Changing pasture management affects how P is recycled through the system and how much P remains in soil pools. Previous studies have investigated the effects of management on P cycling, including long-term SSP application (Nguyen and Goh 1992; Tian et al. 2019), drought (He and Dijkstra 2014; Zhang et al. 2020) and production system (Friesen et al. 1997). However, regenerative agriculture has not been investigated for its effects on P cycling, though others have speculated on the effects (Kaur et al. 2024; Voisin et al. 2024). In this experiment, we quantified changes in the most sensitive pools of P in the soil (microbial biomass P and plant-available P), as well as the movement of P from the soil through grazing animals and its return in dung. This study advances

understanding of P nutrient cycling by demonstrating that conventional assessments of labile P pools may overlook management-driven changes in P dynamics, which were instead expressed through altered biological and animal-mediated recycling pathways.

Soil phosphorus

To understand P cycling, the soil P pools are central to quantifying accumulation or depletion from the system. We quantified three different soil plant-available P tests to determine whether adequate P was available using alternative soil testing approaches. The Mehlich P test extracted the most P (LF; 42 mg P kg⁻¹ and HF; 107 mg P kg⁻¹); the economic optimum in crops has been reported as 51–100 mg P kg⁻¹ (Sims et al. 2002). Our LF treatments were below the economic optimum, averaging 42 mg P kg⁻¹ after spring 2023. However, depending on the location, the critical Mehlich P concentration has been reported to be as low as 30 mg P kg⁻¹ (Zhang et al. 2021). A similar result was also found in the neutral pH extraction of resin P, where there were no differences between regenerative and conventional treatments, only P fertiliser treatment. Resin P is an uncommon soil labile P test in New Zealand and has not been calibrated to pasture systems. However, Sagger et al. (1992) reported an equation resulting from a pot experiment with ryegrass. Using this equation, 90% of maximum yield was a resin P concentration of 61 mg P kg⁻¹. Therefore, like the Mehlich P test, the HF treatments were on average above this threshold of 61 mg P kg⁻¹ at 77 mg P kg⁻¹, while the LF treatments were below at 35 mg P kg⁻¹, indicating deficiency. However, the below optimum soil labile P tests did not impact pasture yield (Watson et al. 2026b). There is no clear explanation of this result, and further research is required to understand the mechanisms involved.

The Olsen P test was the only labile P test to show a difference between regenerative and conventional treatments. In spring 2024, the HF conventional Olsen P was higher than the HF regenerative Olsen P. This was likely due to the application of RPR in the regenerative rather than SSP in that year. RPR requires acidic soil conditions and adequate rainfall to dissolve due to its partial solubility (Bolan and Hedley 1998). In the two weeks between the spring 2024 fertiliser application and soil

sampling, it was unlikely that the RPR would have dissolved into the soil. The tests used in this experiment demonstrated higher P in the plant-available pools of the HF treatments, and little management effect. Collectively, these results demonstrated that labile soil P pools were largely insensitive to management over the duration of this experiment. This was likely driven both buffering of the inorganic P pool by microbial biomass turnover and organic P mineralisation (Frossard et al. 2000), and exchange with inorganic P pools not captured by Mehlich, resin or Olsen P tests. Because we measured plant-available P, we cannot separate the relative contribution of these abiotic pools from the biological mechanisms discussed above. To further understand this result, we also measured biological pools and processes, which may be more sensitive to pasture management than traditional soil tests, which estimate plant-available P.

Microbial biomass is defined as the living component of the soil organic matter, excluding plant roots and soil fauna larger than $5 \times 10^3 \mu\text{m}^3$ (Jenkinson and Ladd 1981). It is widely used as an indicator of soil health because of its relatively short turnover time (<1 year), making it responsive to environmental and management changes that ultimately influence soil organic matter levels (Rice et al. 1997). Microbial biomass P in this experiment peaked at 27 mg P kg⁻¹ in autumn 2025 in the HR treatment, following increases of 35, 91, 12 and 44% for HC, HR, LC and LR respectively since spring 2023. This peak was substantially lower than that reported by Parfitt et al. (2005) in a comparative experiment, in which microbial biomass P was 76 mg P kg⁻¹ (at Olsen P 25 mg P kg⁻¹). As for the larger increases in microbial biomass P in the regenerative treatments, this may be linked to changes in microbial biomass carbon, soil moisture or botanical composition (Watson et al. 2026a). The delayed response of microbial biomass P to fertiliser in this experiment consistent with findings by Chen et al. (2014) who reported microbial biomass P increased with P fertilisation, however Touhami et al. (2022) reported no significant effect of long-term (70 years) P fertilisation on soil microbial biomass P at Winchmore research station. Those results indicated that in a shorter-term experiment like ours, we may see an initial spike in microbial biomass in response to P fertiliser, but the long-term Winchmore site suggests this may not be a long-term effect. The

increase in microbial biomass due to P fertiliser application supports our first hypothesis that fertiliser P application would increase microbial biomass P and is an important finding in this experiment. However, it should be noted that microbial biomass P and labile forms of P may be influenced by soil moisture (Liebisch et al. 2014) and rainfall following sampling may have influenced both forms of P in soil, which is impossible to measure unless sampling is carried out extremely frequently.

Acid phosphatase is excreted by the microbial biomass and plant roots to hydrolyse organic P compounds, releasing inorganic P that can be assimilated (Manzoor et al. 2022; Margalef et al. 2017). Activity was highest overall in autumn 2025, reaching 631 nmol activity g⁻¹ dry soil hr⁻¹ (Fig. 2), and increased 73% on average between spring 2023 and autumn 2025 across all treatments. Treatment differences were generally small and only emerged in the final sampling, when the regenerative treatment showed higher activity than conventional. The lack of P fertiliser response contradicts our hypothesis that P fertiliser would suppress acid phosphatase activity, and contrasts with studies reporting declines in acid phosphatase activity with increasing P inputs (Olander and Vitousek 2000; Touhami et al. 2022; Uwituze et al. 2022). In particular, this was quantified in a global meta-analysis, where phosphatase activity decreased by 20% (Xiao et al. 2018). One explanation is that soil P concentrations in this experiment remained below the threshold required to competitively inhibit phosphatase activity (Manzoor et al. 2022). The lack of P fertiliser response is consistent with findings by Mills et al. (2026), who reported no phosphatase activity difference between treatments at RADE in plots differing to this experiment. However, the extended duration of our experiment aligned with changes in botanical composition, particularly in the regenerative treatment, over time. This change in botanical composition was reported by Watson et al. (2026b) where grass content increased in the regenerative treatment over time. This supports the view that management effects on soil microbial function are mediated through plant community change rather than direct microbial responses to P (Bogdanov et al. 2026). Given the small magnitude and late emergence of this effect, we cautiously interpret this result as a potential early indicator of management effects on P cycling which may manifest in microbial function

before measurable changes in soil P stocks, rather than as established evidence of a treatment effect.

Herbage phosphorus

Variation in herbage P concentrations were observed in this experiment. Herbage P concentrations, along with measurements of animal herbage P intake, can be used to quantify the amount of P removed from the soil by plants. Pre-grazing herbage P concentrations across each treatment and season ranged from 1.4 g P kgDM⁻¹ (LR, autumn 2024) to 4.4 g P kgDM⁻¹ (HC, winter 2024). Similar findings were reported by Rowarth et al. (1988), in a rotationally grazed hill country pasture, herbage P concentration was highest in winter and spring (4.8 g P kgDM⁻¹) and lowest in summer (1.8 g P kgDM⁻¹). Those concentrations were averaged across various P fertiliser rates; however, only one month (September 1983) had a significant effect of P fertiliser application on pasture P concentration in that experiment. A similar marginal response to P fertiliser was observed in this experiment, with the HF treatments resulting in 0.4 g P kgDM⁻¹ higher herbage P concentration than the LF treatments across all years ($P < 0.001$). Our yearly averages support findings by Nguyen and Goh (1992), who reported that in non-stock camp areas herbage P concentrations ranged from 1.6 to 2.7 g P kgDM⁻¹ in the control, from 3.1 to 4.0 g P kgDM⁻¹ in a 188 kg SSP ha⁻¹ treatment and from 3.9 to 5.0 g P kgDM⁻¹ in a 376 kg SSP ha⁻¹ treatment. Compared to our findings, the average concentrations for the LF and HF treatments were 2.7 and 3.0 g P kgDM⁻¹, respectively; both were near the lower end of the 188 kg SSP ha⁻¹ fertiliser treatment reported by Nguyen and Goh (1992). This indicates that both of our fertiliser treatments were comparable to medium fertiliser in that longer-term experiment, in terms of herbage P concentrations.

Herbage P concentrations for meeting plant requirements can vary substantially between plant species. In lucerne plants, Moir et al. (2016) reported that 97% of the maximum yield occurred at a herbage P concentration of 2.9 g P kgDM⁻¹, and for white clover, 3.2 g P kgDM⁻¹. In summer, across all three years, pre-grazing herbage P concentration was below 3.0 g P kgDM⁻¹, which may have limited herbage yield (Fig. S3). This response was likely due to moisture, as soil moisture was negatively linearly

correlated with herbage P concentrations (Fig. S4). The mechanism behind this response was investigated by Bista et al. (2018), who reported that drought reduced herbage P concentration by 39–48%, and this was correlated with reduced P uptake and protein concentrations in the plant. Therefore, this may have occurred in both summer and autumn in this experiment, given the observed lower moisture in this summer-dry environment. In contrast, moisture was abundant in spring (Watson et al. 2026b), and as a result, 40, 71, and 43% of the mean annual yield across all treatments occurred in spring 2022, spring 2023, and spring 2024, respectively (Watson et al. 2026b). In the first two springs, there was no difference in animal P intake between treatments, suggesting that each treatment was not moisture limited. The importance of soil moisture on P cycling in this environment was shown in a PCA where soil moisture had the largest loading on PC1, which accounted for 31% of the variance in the data (Fig. 6). This supports findings by Moir et al. (2000) where plant responses to P are often driven by moisture in dryland systems.

Dung phosphorus returns

Dung P concentrations were only influenced by P fertiliser late in the experiment (Year 3). Dung P returns indicate P recycling in each treatment, which is important for effective and efficient P cycling in the system. Dung P returns in this experiment were calculated using a combination of measured dung P concentrations and estimated dung excretion quantities using pasture quality and animal intake (Eq. 1). Dung P concentrations in spring and autumn over two years were, on average, higher in the HF treatment than the LF (9.0 vs. 7.8 g P kgDM⁻¹). Rowarth et al. (1988) also reported that increased fertiliser P application increased dung P concentrations and Williams and Haynes (1992) reported dung P concentrations of 4.8, 7.4 and 11 g P kgDM⁻¹ in the control, 188 and 376 kg SSP ha⁻¹ treatments, respectively. Interestingly, the mean dung P concentration in our LF treatment was similar to the 7.4 g P kgDM⁻¹ reported by Williams and Haynes (1992) in the medium fertiliser treatment (188 kg SSP ha⁻¹). This suggests that, in this experiment, the LF dung P concentrations were comparable to medium fertiliser inputs of 188 kg SSP ha⁻¹, which was not considered to be low fertility at that experiment. The significant effect of P fertiliser on

dung P concentrations was observed only in Year 3, likely because herbage P concentrations also differed significantly between fertiliser treatments in that year. This effect represents a potential lag between soil and plant P, whereby soil plant available P concentrations were low, and plant P concentrations were higher than expected for the given soil P fertility status. Only after three years did the soil labile P concentrations reflect the trend in herbage P concentrations.

Dung P concentrations were higher in the conventional treatment than in the regenerative treatment (Fig. 4). This was likely due to differences in pasture quality. Arnuti et al. (2020) reported that dung sampled at the pasture vegetative stage was 16% higher in P than dung collected at the post-flowering stage. In this experiment, the regenerative pasture grasses flowered in late summer and autumn due to longer grazing rotations, which may have reduced the dung P concentrations of that treatment. The lucerne in the conventional treatment was frequently topped post-grazing and had shorter grazing rotations, resulting in continued vegetative growth with higher P concentrations than reproductive pastures. The PCA analysis revealed that the points representing different treatments were relatively evenly distributed across the ordination space, and the ellipses for different seasons were clearly separated, particularly for autumn samplings. This demonstrated that seasonal variation had an impact on key P cycling components as well as treatment in this experiment. The fluctuations in pasture quality by season could explain the seasonal variations in dung P concentrations that were also reported by Rowarth et al. (1988) where they reported dung P concentrations ranging from ~4 to 14 g P kgDM⁻¹. In this study, the range was 4.9 to 12 g P kgDM⁻¹ (Fig. 5). This indicates that although dung concentration seems to vary seasonally across different experiments, the concentrations are relatively consistent.

The calculated dung returns were higher in the conventional than the regenerative in the first two years (by 7 and 4 kg P ha⁻¹ in Year 1 and Year 2, respectively), but higher in the regenerative in Year 3 (by 3 kg P ha⁻¹). This may in part be attributed to a change in pasture management between years. Mowing was implemented post-grazing in the regenerative pastures in Year 3 to manage excessive seed head production of *Dactylis glomerata* (cocksfoot) and *Cichorium intybus* (chicory), as recommended by the

independent technical advisory group of regenerative farmers. However, because of the mowing, the pasture stayed in a vegetative state, and the sheep likely consumed more vegetative pasture than reproductive pasture in Year 3 than in the previous two years.

Dung P returns were based on the P concentration of dung samples collected during the main pasture growth period of the year (spring and summer). To improve this measurement, dung samples could be collected in all seasons. The dung returns were also calculated using ewe energy requirements and average pasture quality. This calculation resulted in higher daily dung DM per sheep for the regenerative treatment due to lower pasture quality/digestibility (Eq. 2). To know the actual quantity of dung produced by a sheep, the dung would need to be directly measured, either using feeding crates or harnesses to capture all dung produced. While other experiments have measured dung outputs in pasture (Hurley et al. 2021), variation in pasture quality, pasture type and animal age and breed makes it challenging to compare those experiments to this one. However, higher dung returns in the regenerative treatments in Year 3 indicated that P recycling is greater compared to the conventional treatments by the end of this experiment.

However, dung P returns were calculated based on measured dung P concentrations and pasture quality estimates. Therefore, there are limitations to this measurement. These include difficulty to obtain representative samples without sampling after every grazing event. Also, the dung DM returns are based on single pasture quality values reported by Robb et al. (2025), where in reality pasture quality likely fluctuates during the season (Hill et al. 2009). Because the calculations used in this experiment use single value assumptions, it should be noted that there is uncertainty in the dung returns calculation. This is an area of potential future research where dung quantity and quality can be measured throughout the seasons to accurately capture the P (and other nutrient) contributions of contrasting management systems.

Phosphorus removal

To estimate P removal in each treatment, the P removed in products was quantified. The P removed in meat products was low in this experiment and occurred only in Years 2 and 3, with a maximum loss of 2.6 kg P ha⁻¹ yr⁻¹ (HC, Year 3). Comparatively, in

a conventional grazed fertiliser trial at Winchmore, New Zealand, product removal was reported as 0.6 kg P ha⁻¹ yr⁻¹ in the control and up to 1.3 kg P ha⁻¹ yr⁻¹ in the 376 kg P ha⁻¹ yr⁻¹ fertiliser application treatment (Tian et al. 2019). The product removals in Year 2 of this experiment were similar to those reported in Tian et al. (2019) with less than 2 kg P ha⁻¹ removed across all treatments (Table S1). The removal of P from wool was included among the meat product losses in this experiment, as it was less than 0.05 kg P ha⁻¹ yr⁻¹ for each treatment. To compare with another experiment, the animal product outputs of P on a mixed cropping farm were 6 kg P ha⁻¹ and 7 kg P ha⁻¹ for conventional and organic farm systems, respectively, in the same soil type as this experiment (Nguyen et al. 1995). Those findings were much higher than those reported in this experiment and demonstrate the variability of animal product P removals across systems, depending on stocking rate, feed availability, and P intake. In this experiment, the animal product export of P was a minor pathway of P loss relative to fertiliser inputs and recycling fluxes.

Animal transfers were deemed negligible due to the small plot sizes (29 × 30 m), flat terrain and relatively high stocking rate (10–14 sheep ha⁻¹). Animal transfers in dryland systems are often related to slope and aspect in more extensive systems with lower pasture yields and stocking rates (Saggar et al. 1990b; Taylor et al. 1987). In this experiment, potential transfer loss may have occurred along fence lines; further investigation is warranted if substantial amounts of dung are deposited there, particularly where one plot borders another. However, this was not measured in this experiment. There was also potential for minor losses due to sheep defecation between shifts in laneways and in the yard. While there are multiple P loss pathways in agricultural systems, of the losses measured these were minor (< 3 kg P ha⁻¹) compared to the fertiliser inputs (> 50 kg P ha⁻¹ for high fertiliser). This result suggests inefficient fertiliser use. However, some caution must be used in interpreting this result. For example, this study presents the first three year's data of a longer-term experiment. Over time, with more data, it is possible that P balances and efficiencies may change in these systems. Further, obvious knowledge gaps have arisen about how these pastures are accessing soil P, especially in what we would currently define as being low labile P soils.

With these limitations in mind, this research still provides valuable and novel insights into P cycling under alternative management of a dryland pasture system. Together, these findings indicate that regenerative management potentially alters acid phosphatase activity and recycling pathways, such as dung returns, before any measurable changes occur in plant-available soil P pools like Olsen P. In shorter-term studies, such as this one, these results highlight the importance of incorporating biological indicators into P cycling measurements to detect early management impacts.

Conclusions

This quantification of P inputs, pools and outputs has provided new information on P cycling in regenerative and conventional grazing systems with contrasting fertiliser inputs. In this dryland environment, soil moisture was a key driver of herbage P uptake and consequently animal P intake and excretion. The regenerative treatment had contrasting pasture composition to the conventional which therefore influenced dung P recycling. Over time, effects of P fertiliser input became more evident, and by Year 3, differences were reflected in dung P returns and soil P status. Acid phosphatase activity was increased on average in the regenerative compared to conventional treatments in the final sampling. Whereas traditional chemical P tests, such as Olsen P were not influenced by management system, only P fertiliser inputs. This result suggests the microbial community may respond to management change and may potentially serve as a more sensitive indicator of P availability than Olsen P. However, the difference in acid phosphatase activity may also reflect the contrasting botanical composition between the regenerative and conventional treatments rather than management, and further research is required to disentangle these effects. Phosphorus removal in animal products was low across systems, potentially leading to P surplus in each system. Continued investigation is required to capture the trajectory of soil P pools over longer time periods and to quantify the longer-term influence of contrasting fertiliser inputs and management systems.

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Author contribution KW, conceptualisation, data curation, formal analysis, visualisation, writing – original draft, writing – review & editing. JM, conceptualisation, supervision, funding acquisition, writing – review & editing. LC, supervision, writing – review & editing. CA, supervision, writing – review & editing. AB, supervision, project administration.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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